

Chapter 05 Homework — Tests for irreducibility

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Problem 05.4. Suppose that K is a field and that f and g are relatively prime in $K[x]$. Show that $f - yg$ is irreducible in $K(y)[x]$.

Problem 05.6. Suppose that R is an integral domain and that

$$f = f_0 + f_1x + \cdots + f_nx^n \in R[x]$$

has the property that $f_0, \dots, f_n$ are relatively prime. Suppose that p is a prime in R , and that $p \mid f_i$ for $1 \leq i \leq n$, while p does not divide f_0 and p^2 does not divide f_n . Show that f is irreducible in $R[x]$.

Proof. Reduce the coefficients of f modulo p . Since $p \mid f_i$ for $1 \leq i \leq n$, we get $\bar{f} = \bar{f}_0$, which is of degree zero, as $p \nmid f_0$.

Suppose $f = gh$ with $g, h \in R[x]$. Then $\bar{f} = \bar{g}\bar{h}$, and since (p) is prime, $R/(p)$ is an integral domain, so degrees add and both $\bar{g}$ and $\bar{h}$ must be of degree zero.

Case 1: g and h both of positive degree. Write $k = \deg g$ and $m = \deg h$, so $f_n = g_k h_m$. Since $\bar{g}$ and $\bar{h}$ are constants, the leading coefficients vanish mod p : $p \mid g_k$ and $p \mid h_m$. Hence $p^2 \mid f_n$, a contradiction.

Case 2: g or h of degree zero. Without loss of generality $\deg h = \deg f$ and $g(x) = c \in R$. Then $f_i = c h_i$ for all i , so $c \mid f_i$ for every i . Since the f_i are relatively prime, c must be a unit, and so f is irreducible in this case too. $\square$

Problem 05.7. Show that if p is a prime number then $x^n - p$ is irreducible in $\mathbb{Q}[x]$.

Problem 05.8. Let A denote the field of real numbers which are algebraic over $\mathbb{Q}$. Show that $[A : \mathbb{Q}] = \infty$.

Problem 05.10. Show that $x^5 - 4x + 2$ and $x^4 - 4x + 2$ are irreducible over $\mathbb{Q}(i)$.